

Evolutionary Game Theory and Imitation Dynamics

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Population Games

Consider a large population of $N_t \gg 1$ identical agents.

Strategy revision in continuous time: agents are paired randomly to play symmetric two-player games.

The fraction of agents choosing action $a \in A$ at time t is

$$x_a(t) = \frac{n_a(t)}{\sum_{a' \in A} n_{a'}(t)} = \frac{n_a(t)}{N_t}$$

The expected utility of action a is

$$\bar{u}_a(x) = \sum_{a'} x_{a'}(t) u(a, a')$$

The expected average utility over the population is

$$\bar{u}(x) = \sum_a x_a(t) \bar{u}_a(x)$$

Population Games

The number of agents changes according to an **evolutionary principle** (average payoff $\iff$ *fitness*):

$$\dot{n}_a(t) = n_a(t) \bar{u}_a(x)$$

i.e. agents playing strategy a generate a number of copies of themselves proportional to their fitness. The fractions undergo the **Replicator Equations**

$$\frac{d}{dt}x_a(t) = x_a(t) (\bar{u}_a(x) - \bar{u}(x))$$

For two strategies $a \in \{A, B\}$ with $x = n_A(t) / (n_A(t) + n_B(t))$

$$\frac{dx}{dt} = x(1-x) (\bar{u}_A(x) - \bar{u}_B(x))$$

Population Games

A **population game** is defined by a single unit-mass population with S pure strategies and a population state defined by the shares $x(t) \in X$ with $x_a(t) \geq 0$ and $\sum_{a=1}^S x_a(t) = 1$.

The *revision protocol* is generally defined by a function $\rho_{aa'}(x, u)$, so that

$$\dot{x}_a = \sum_{a'} x_{a'} \rho_{a'a}(x, u) - x_a \sum_{a'} \rho_{aa'}(x, u)$$

An **imitative** protocol is such that $\rho_{aa'}(x, u) = x_{a'} r_{aa'}(x, u)$, that gives

$$\dot{x}_a = x_a \sum_{a'} x_{a'} (r_{a'a}(x, u) - r_{aa'}(x, u))$$

Replicator equations are obtained for $r_{aa'}(x, u) = \max(0, \bar{u}_{a'}(x) - \bar{u}_a(x))$.

Population Games

Nash Equilibrium: a population state x^* such that either $x_a^* = 0$ or $x_a^* > 0$ implies $\bar{u}_a(x^*) \geq \bar{u}_{a'}(x^*)$, $\forall a' \in A$.

A NE is a fixed point of the RE, but not all fixed points of RE are NE.
Stable fixed points x^* of RE are NEs.

Consider strategies of symmetric two-player games that are resistant to invasion by other strategies.

Definition

The strategy s is **evolutionary stable** iff $\exists \epsilon_0 > 0$ s.t. $\forall \epsilon < \epsilon_0$ and $\forall s' \neq s$,

$$(1 - \epsilon) u(s, s) + \epsilon u(s, s') > (1 - \epsilon) u(s', s) + \epsilon u(s', s').$$

equivalently, s is such that

- ① $u(s, s) > u(s', s)$
- ② or if $u(s, s) = u(s', s)$, then $u(s, s') > u(s', s')$.

Population Games

Is there a relation between evolutionary stability and NE in population games?

Proposition

In a population game G , if a population state x^ corresponds to an evolutionary stable strategy (ESS) of a two-player game, then x^* is a NE of G . If x^* is a strict NE of G , then x^* is an ESS.*

Is there a relation between evolutionary stability and stability of RE?

Proposition

In a population game G , if a population state x^ corresponds to an ESS of a two-player game, then x^* is an asymptotically stable state of the replicator equations for G .*

For a **finite population** of agents, it is reasonable to consider stochastic effects, using formulations that are equivalent to those used in population genetics (Moran and Wright-Fisher process).

Moran process:

- fixed population size, N individuals labeled by a strategy among the available ones
- at every time step an agent is randomly selected for reproduction with a probability proportional to the relative abundance n_a/N of their strategy a and to the relative value $\bar{u}_a/\bar{u}$ of the expected utility (relative fitness).
- reproduction takes place at the expenses of another randomly selected individual.

Consider two-strategies and set $n = n_A$ and $x = n_A/N$.

The transition probabilities (when N games are played) are

$$\begin{aligned}W_n^+(x) &= Nx(1-x) \frac{\bar{u}_A(x)}{\bar{u}(x)} \\W_n^-(x) &= Nx(1-x) \frac{\bar{u}_B(x)}{\bar{u}(x)}\end{aligned}$$

Master Equation (in continuous time)

$$\partial_t P(n, t) = W_{n+1}^- P(n+1, t) + W_{n-1}^+ P(n-1, t) - (W_n^+ + W_n^-) P(n, t)$$

Stochastic Dynamics

Defining

$$\mathbb{E}_t[n] = \sum_n n P(n, t)$$

its dynamics involves higher-order moments (not closed)

$$\frac{d}{dt} \mathbb{E}_t[n] = \mathbb{E}_t[W_n^+] - \mathbb{E}_t[W_n^-]$$

Using a **mean-field approximation** (factorizing higher-order moments)

$$\bar{x}(t) = \frac{1}{N} \mathbb{E}_t[n], \quad \frac{\mathbb{E}_t[n^2]}{N^2} \approx \bar{x}^2, \dots$$

we get the (rescaled) *replicator equations*

$$\frac{d}{dt} \bar{x} = \bar{x} (1 - \bar{x}) \frac{\bar{u}_A(x) - \bar{u}_B(x)}{\bar{u}(x)}.$$

Performing a Kramers-Moyal system-size expansion,

$$W_{n+1}^- = W_{N(x+\frac{1}{N})}^- = N \left(w_x^- + \frac{1}{N} \partial_x w_x^- + \frac{1}{2N^2} \partial_x^2 w_x^- + \dots \right)$$

$$W_{n-1}^+ = W_{N(x-\frac{1}{N})}^+ = N \left(w_x^+ - \frac{1}{N} \partial_x w_x^+ + \frac{1}{2N^2} \partial_x^2 w_x^+ + \dots \right)$$

and

$$P(n+1, t) = \frac{1}{N} \left(p(x, t) + \frac{1}{N} \partial_x p(x, t) + \frac{1}{2N^2} \partial_x^2 p(x, t) + \dots \right)$$

$$P(n-1, t) = \frac{1}{N} \left(p(x, t) - \frac{1}{N} \partial_x p(x, t) + \frac{1}{2N^2} \partial_x^2 p(x, t) + \dots \right)$$

Keeping up to $1/N$ corrections in the expansion of the Master equation, we obtain the **Fokker-Planck equation** (diffusive approximation)

$$\partial_t p(x, t) = -\partial_x [M(x) p(x, t)] + \frac{1}{2} \partial_x^2 [D(x) p(x, t)]$$

with

$$M(x) = w_x^+ - w_x^- = x(1-x) \frac{\bar{u}_A(x) - \bar{u}_B(x)}{\bar{u}(x)}$$
$$D(x) = \frac{w_x^+ + w_x^-}{N} = \frac{1}{N} x(1-x) \frac{\bar{u}_A(x) + \bar{u}_B(x)}{\bar{u}(x)}$$

The stochastic dynamics admits two absorbing states at $x = 0, 1$.

Eventually, for any large but finite N , stochastic fluctuations will drive the system in the absorbing states (extinction of one strategy/trait).

Stochastic Dynamics

Quasi-stationary assumption

$$P_{\text{st}}(x) = \mathcal{N}^{-1} e^{-V(x)}$$

and considering no currents

$$\partial_t p(x, t) = \partial_x J(x, t) = 0$$

we have

$$\begin{aligned} 0 &= -[M(x) p_{\text{st}}(x)] + \frac{1}{2} \partial_x [D(x) p_{\text{st}}(x, t)] \\ &= -[M(x) e^{-V(x)}] + \frac{1}{2} [D'(x) e^{-V(x)} - V'(x) D(x) e^{-V(x)}] \end{aligned}$$

that gives

$$V(x) = \int^x dx \frac{-2M(x) + D'(x)}{D(x)}$$

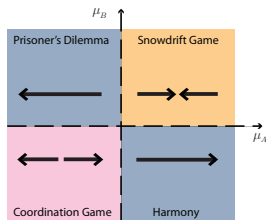
This potential describes the long time behavior of the population.

Stochastic Dynamics

Given the utility bimatrix

	A	B
A	a, a	b, c
B	c, b	d, d

and defining $\mu_A = b - d$ and $\mu_B = c - a$ we get the overall behavior

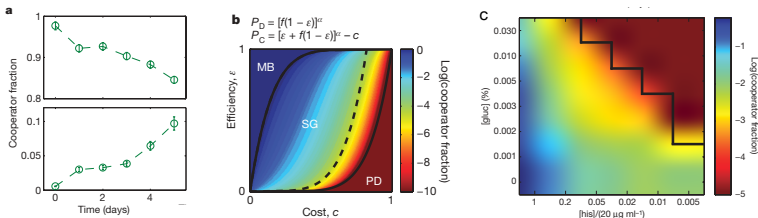


Game	control parameters	stable fixed points
prisoner's dilemma	$\mu_A < 0 ; \mu_B > 0$	0
snowdrift	$\mu_A > 0 ; \mu_B > 0$	$\mu_A / (\mu_A + \mu_B)$
coordination	$\mu_A < 0 ; \mu_B < 0$	0, 1
harmony	$\mu_A > 0 ; \mu_B < 0$	1

Stochasticity perturbs the internal fixed points driving the population to the extremal ones.

Microbial games

Yeast (*Sac. cerev.*) secrete invertase (enzyme) to hydrolyze sucrose into glucose and fructose, but about 99% of the resulting sugars diffuse away, so invertase is a *public good*: it is costly to the producer but benefits neighboring cells.¹

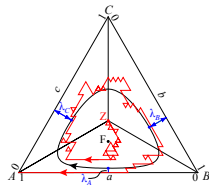
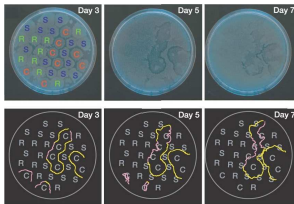
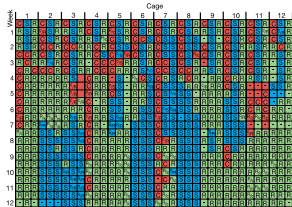


Well-mixed cultures: cheaters (mutant lacking the invertase gene) can invade cooperator populations, and viceversa (coexistence like in hawk–dove games). Yeast cooperate when competing with cheaters (low glucose) and defect when surrounded by cooperators (high glucose)

Microbial games

Cyclic dominance is observed in bacteria:

Colicins are narrow-spectrum antibiotics produced by and active against *Escherichia coli* and its close relatives. Strains that produce colicins (C) kill sensitive (S) strains, which outcompete resistant (R) strains, which outcompete C strains.²



²Left: B. Kirkup et al. Nature (2004); Centre: B. Kerr et al, Nature (2002).

Voter model

N agents have two possible opinions $a \in \{+1, -1\}$. An agent updates the opinion imitating another randomly selected agent.

Opinion numbers n_+ and $n_- = N - n_+$ change with rates

$$w_{n_+}^+(n_+) = w_{n_+}^-(n_+) = \frac{n_+(N - n_+)}{N(N - 1)}$$

The Fokker-Planck equation is

$$\partial_t p(m, t) = \frac{1}{2} \partial_m^2 \left[\frac{1 - m^2}{N} p(m, t) \right]$$

where the “magnetization” $m = (n_+ - n_-) / N$ is conserved

$$\frac{d}{dt} \mathbb{E}_t[m] = 0$$

Voter model

The dynamics is driven by the presence of active links

$$\rho(t) = \frac{2n_+n_-}{N(N-1)} \approx \frac{1 - m^2(t)}{2}$$

Their density change under single-opinion flip

$$\Delta\rho_{+\rightarrow-} = \frac{(n_+ - 1)(n_- + 1) - n_+n_-}{N(N-1)/2} = \frac{2n_+ - N - 1}{N(N-1)/2}$$

$$\Delta\rho_{-\rightarrow+} = \frac{(n_- - 1)(n_+ + 1) - n_+n_-}{N(N-1)/2} = \frac{2n_- - N - 1}{N(N-1)/2}$$

with rate $\lambda_{+\rightarrow-} = \lambda_{-\rightarrow+} = \frac{n_+n_-}{N-1}$ per system's update.

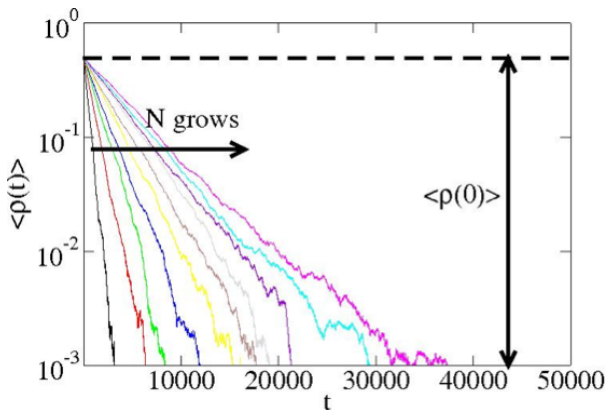
It gives the rate equation

$$\begin{aligned} \frac{d}{dt}\mathbb{E}_t[\rho] &= \mathbb{E}_t[\lambda_{+\rightarrow-}\Delta\rho_{+\rightarrow-} + \lambda_{-\rightarrow+}\Delta\rho_{-\rightarrow+}] \\ &= -\frac{2}{N-1}\mathbb{E}_t[\rho] \end{aligned}$$

Voter model

$$\mathbb{E}_t[\rho] \approx e^{-t/\tau(N)},$$

where the average time to reach consensus $\tau(N) \propto N$.



Voter model

Average time $\tau(n)$ to reach consensus when starting with $n_+ = n$ agents.

With transition rates $w_n^{+/-}(n)$, we get the backward equation

$$\tau(n) = 1 + w_n^+(n) \tau(n+1) + w_n^-(n) \tau(n-1) + (1 - w_n^+(n) - w_n^-(n)) \tau(n)$$

with boundary conditions $\tau(0) = \tau(N) = 0$.

In the large N limit, it gives

$$\frac{1-m^2}{2N} \frac{\partial^2}{\partial m^2} \tau(m) = -1$$

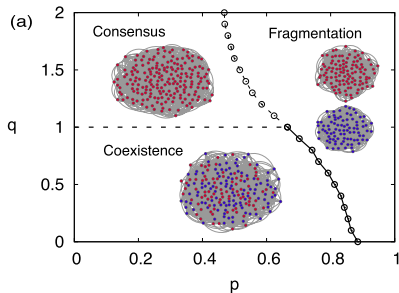
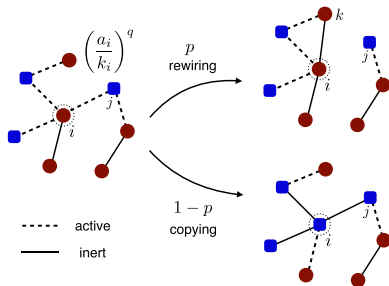
that is

$$\tau(m) = -N \left[(1+m) \log \frac{1+m}{2} + (1-m) \log \frac{1-m}{2} \right]$$

Voter model

Fragmentation transition in coevolving networks:³

non-linear voter model + possible rewiring when disagreement lead to fragmentation



³ Min et al., *Sci. Rep.* (2017)

Language Games

Language competition⁴: a system of two competing languages A, B , in which the attractiveness of a language increases with both its number of speakers and its perceived status

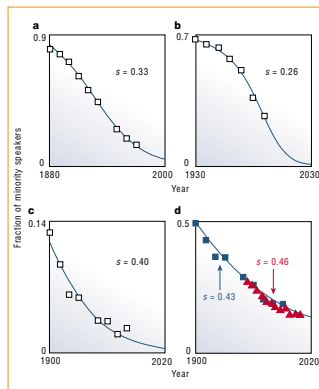
x = fraction of language-A speakers

$$\frac{dx}{dt} \propto s(1-x)x^a - (1-s)x(1-x)^a$$

Variable s is the status. The exponent is close to 1 ($a \approx 1.3$ across data).

For $s = 1/2$ and $a = 1$, the Voter model is recovered.

Figure 1 The dynamics of language death. Symbols show the proportions of speakers over time of: a, Scottish Gaelic in Sutherland, Scotland⁸; b, Quechua in Huanuco, Peru; c, Welsh in Monmouthshire, Wales¹⁰; d, Welsh in all of Wales, from historical data¹⁰ (blue) and a single modern census¹¹ (red). Fitted curves show solutions of the model in equation (1), with parameters c , s , a and $x(0)$ estimated by least absolute-values regression. Where possible, data were obtained from several population censuses collected over a long timespan; otherwise, a single recent census with age-structured data was used (although errors are introduced, the size of which are reflected in the differing fits in d). Using the fraction of Catholic masses offered in Quechua in Peru as an indicator, we reconstructed an approximate history of the language's decline.



⁴Abrams & Strogatz, Nature (2002)

Bilingualism: from the case $s = 1/2$, $a = 1$, add a bilingual state⁵

$$\begin{aligned}\frac{dx_A}{dt} &= \frac{1}{2} (x_A x_{AB} + x_{AB}^2 - x_A x_B) \\ \frac{dx_B}{dt} &= \frac{1}{2} (x_B x_{AB} + x_{AB}^2 - x_A x_B) \\ \frac{dx_{AB}}{dt} &= \frac{1}{2} (2x_A x_B - (x_A + x_B) x_{AB} - 2x_{AB}^2)\end{aligned}$$

Three fixed points depending on initial conditions (and stochasticity):
 $x_A = 1$, $x_B = 1$ (stable) and $x_A = x_B = 2x_{AB} = 0.4$ (unstable).

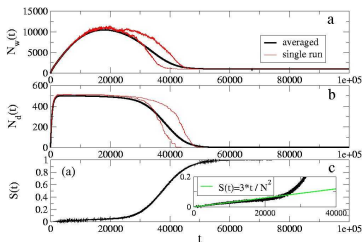
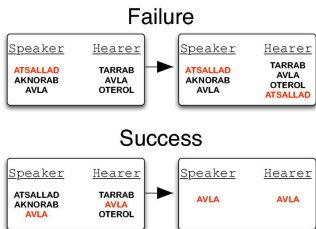
Coexistence becomes a long-lived metastable state on evolving networks, where bilinguals use minoritarian language to speak among themselves and the majority language for communications with the others.⁶

⁵X. Castellò et al., *New J Phys* (2006)

⁶A. Carro et al., *New J Phys* (2016)

Language Games

Naming game⁷: each player is characterized by an inventory of known word-object associations. Inventories are initially empty. Players interact pairwise (speaker-hearer pair): the speaker selects a word; if the hearer does not know it, she includes it in his inventory (left-top); otherwise both agents erase their inventories only keeping the winning word (left-bottom).

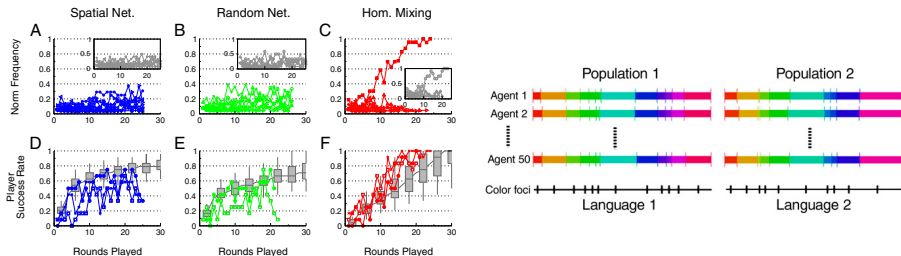


After a transient, **global consensus emerges**, i.e., every agent has the same word for the same object.

⁷L. Steels, *Artif Life* (1995); A. Baronchelli et al. *JSTAT* (2006)

Language Games

Experimental validations: emergence of conventions to name objects (left)⁸; emergence of color categories on a continuous spectrum (right)⁹



⁸ D. Centola et al., *PNAS* (2015)

⁹ A. Baronchelli et al., *PNAS* (2009)